

Physical Edge-Root Matching

The Length-Four Temporal Clock Resolves a Genuine Spectral Quartet

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Abstract

RH-192 shows that a rank-four temporal packet cannot be the complete Riesz shell of the full Frobenius left-multiplication operator. RH-193 leaves open a sharper possibility: the packet may resolve four source-excited spectral modes in the cyclic restriction. This paper tests that possibility at the only surviving physical anchor, $\sigma = 0.01, L = 4$.

The twelve RH-185 windows passing the frozen two-sided 0.10 residual gate contain 48 packet roots. Direct eigendecomposition of the two base physical operators finds a distinct nearest eigenvalue for every root. Each predeclared RH-190 root disc contains exactly one base eigenvalue, the maximum matching error is below 1.3×10^{-3} , and all accepted windows on a given side select the same four physical modes. Thus there are eight unique modes across the left and right channels, not 48 unrelated fits.

For a matched simple eigenvalue with spectral projector P_λ , the source and observation states are $P_\lambda S$ and $P_\lambda^* O^*$. Every one has nonzero residue $\text{tr}(OP_\lambda S)$. The temporal four-space is close to the canonical span of these source modes: the maximum principal-angle sine is between 0.035 and 0.098 on the right packet and between 0.038 and 0.113 on the left packet.

This is a finite floating-point discovery, not an interval spectral enclosure or an all-level quartet theorem. It nevertheless changes the frontier: the local clock did not manufacture a root grid; it approximated a genuine source-observable physical edge quartet.

1 Question after the multiplicity correction

The full matrix state at $\sigma = 0.01$ has width $m = 256$. Hence one simple base eigenvalue produces ambient Frobenius Riesz rank 256, and a literal rank-one root shell is impossible [1]. The source-cyclic restriction generated by the single matrix S , however, can contain one direction from that repeated eigenspace [2].

There are therefore two sharply different explanations of the RH-185 local candidate:

1. the four compressed roots are Ritz artifacts with no nearby physical spectrum;
2. they are approximations to four actual base eigenvalues, viewed through source-excited matrix directions.

This paper performs the direct finite test.

2 Frozen data contract

No window or contour is selected after inspecting the base spectrum. The inputs are frozen by RH-185 and RH-190:

- scale $\sigma = 0.01$;
- packet length $L = 4$;
- starts 14–18 on the left and 12–18 on the right;
- two-sided relative residual threshold 0.10;
- root-disc radius $0.4\rho\sin(\pi/4)$, where ρ is the source-cycle radius of that window.

The base dimensions are 512 on the left and 256 on the right. Both matrix sources have 256 columns.

For each accepted window we reconstruct the balanced right and left frames V, W , the compressed matrix

$$K = W^* \mathcal{L}_A V, \quad (1)$$

and its four eigenvalues. We then compute the complete floating spectrum of the base A and solve a distinct nearest-neighbor assignment.

3 Root matching and contour count

Let $\kappa_1, \dots, \kappa_4$ be the packet roots and let $\lambda_1, \dots, \lambda_4$ be their distinct matched base eigenvalues. The matching error is

$$\epsilon_{\text{match}} = \max_j |\kappa_j - \lambda_j|. \quad (2)$$

Proposition 3.1 (Finite matching result). *Across the twelve accepted windows:*

1. all 48 packet roots have distinct matched base eigenvalues;
2. every frozen root disc contains exactly one base eigenvalue;
3. $\epsilon_{\text{match}} < 1.3 \times 10^{-3}$;
4. all windows on one side select the same four base modes.

The final clause is important. Window motion does not chase different eigenvalues. It improves the approximation to one stable edge quartet.

Representative physical eigenvalues have the geometry

$$-0.495 \pm 0.620i, \quad 0.474 \pm 0.557i, \quad (3)$$

with small left/right discretization differences. Their moduli form two nearby radii, approximately 0.793 and 0.731. This explains why the compressed roots are close to, but not exactly on, one circle.

4 Why the result is compatible with RH-192

For a root disc containing one simple base eigenvalue,

$$N_A = 1, \quad N_{\mathcal{L}_A} = 256. \quad (4)$$

The temporal packet count is one. It agrees with the base count and with a source-cyclic count, but not with the full Frobenius count. The finite match therefore validates the spectral location while confirming the state-type correction.

Equivalently, if P_λ is the rank-one base spectral projector, the full ambient range is

$$\{P_\lambda X : X \in \mathbb{C}^{n \times 256}\}, \quad (5)$$

of dimension 256. The source orbit selects the single matrix $P_\lambda S$ from this range.

5 Source-observation residues

Let v_λ, w_λ be right and left eigenvectors normalized by $w_\lambda^* v_\lambda = 1$, so $P_\lambda = v_\lambda w_\lambda^*$. Define

$$X_\lambda = P_\lambda S, \quad Y_\lambda = P_\lambda^* O^*. \quad (6)$$

Their Frobenius pairing is

$$\langle Y_\lambda, X_\lambda \rangle_F = \text{tr}(O P_\lambda S) =: r_\lambda. \quad (7)$$

The same scalar is the residue at λ of

$$g(z) = \text{tr}(O(zI - A)^{-1}S). \quad (8)$$

Every matched mode has $r_\lambda \neq 0$ at floating precision. The minimum recorded residue modulus across repeated window records is greater than 1.1×10^{-2} . Therefore all four modes are both source-excited and observation-visible. RH-195 proves the exact projector/residue theorem used here without relying on diagonalizability away from the selected simple modes.

6 Canonical spectral subspaces

For one side, define

$$E_R = \text{span}\{X_{\lambda_1}, \dots, X_{\lambda_4}\}, \quad E_L = \text{span}\{Y_{\lambda_1}, \dots, Y_{\lambda_4}\}. \quad (9)$$

These spaces are exactly invariant under $\mathcal{L}_A$ and $\mathcal{L}_A^*$, respectively. In contrast, the temporal spaces $T_R(t), T_L(t)$ are approximate invariant Krylov windows.

For equal-dimensional subspaces E, F , write

$$\text{gap}(E, F) = \sin \theta_{\max}(E, F), \quad (10)$$

where $\theta_{\max}$ is the largest principal angle.

Proposition 6.1 (Finite subspace alignment). *For the accepted windows,*

$$0.0348 < \text{gap}(T_R, E_R) < 0.0978, \quad (11)$$

$$0.0384 < \text{gap}(T_L, E_L) < 0.1133. \quad (12)$$

The smallest gaps occur near the latest accepted windows.

The alignment is stronger information than eigenvalue proximity alone. It shows that the four-dimensional temporal state is approaching the physical source and observation mode spans, not merely reproducing four scalar roots through an ill-conditioned compression.

7 Projector and transversality diagnostics

The selected base eigenvalues are nonnormal modes. Their spectral projector norms are finite but not near one: the maximum is about 17.7. The normalized source-observation overlap

$$\gamma_\lambda = \frac{|r_\lambda|}{\|X_\lambda\|_F \|Y_\lambda\|_F} \quad (13)$$

ranges down to approximately 1.13×10^{-3} on the right channel and 5.46×10^{-3} on the left. Thus the canonical packet is genuinely oblique. The poor cross-angle conditioning seen in RH-185 is partly physical, not solely a temporal-window artifact.

This observation also prevents an overstatement. Root matching does not automatically provide robust all-level coordinates. Residues and transversality must be controlled under refinement.

8 Determinant and trace comparison

For the matched quartet define the exact finite spectral polynomial

$$d_E(z) = \prod_{j=1}^4 (z - \lambda_j). \quad (14)$$

The temporal compression gives $d_T(z) = \det(zI - K)$. We store the determinant error and the trace-power differences

$$\text{tr } K^q - \sum_{j=1}^4 \lambda_j^q, \quad 1 \leq q \leq 8. \quad (15)$$

These data quantify more than pointwise root matching and will be used in RH-199 to formulate a source-channel determinant and trace ledger.

The errors decrease strongly as the temporal subspaces align, but no asymptotic rate is inferred from seven starts. The data are a finite calibration, not a convergence theorem.

9 Numerical reliability and boundary

The base eigendecompositions are recomputed once per side using a standard complex nonsymmetric solver. Matching is distinct and contour counts are performed against the complete returned spectrum. The minimum distance of any computed eigenvalue from a frozen contour is stored to expose contour fragility.

What is not supplied is an interval enclosure of the matrix entries, eigenvalues, or contour winding number. Therefore “one eigenvalue per disc” is a floating finite statement. The matching errors are roughly two orders of magnitude smaller than the contour radii, which makes the signal clear, but it is not a computer-assisted proof.

Nor is the quartet yet intrinsic. It is found by comparing a late temporal window with the complete physical spectrum. A Gate-A construction must select and transport the relevant modes without target-dependent matching.

10 Updated frontier

The local logical chain is now

temporal packet [approximate, source-generated]
→ one genuine physical edge quartet [finite]
→ source-observation spectral channels [next]
→ canonical packet and transport [open].

This is a meaningful advance within Gate A. It does not address the completion to a unitary/scattering object, a self-adjoint generator, the $T \log T$ law, von Mangoldt traces, zeta divisors, Hilbert–Pólya, or RH.

References

- [1] Liang Wang. The frobenius left-multiplication obstruction, 2026. Prime Dynamics Theory Program, RH-192.
- [2] Liang Wang. Source-cyclic spectral quotients, 2026. Prime Dynamics Theory Program, RH-193.